

HW 2 — Solutions and Marking Scheme

ID-5-02 (TO) · Mathematical Foundations of Diffusion Models · Monsoon 2026

Worked solutions and marking scheme, released for student reference.

Total: 50 points (5 problems $\times$ 10). **Graded on correctness** — unlike HW 1. The per-part marks below are the marking scheme, not a formality.

Problem 1 — Beta mean and variance (10)

Setup: germination rate varying across seed lots; Beta(α, β) on $[0, 1]$.

Throughout, $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha + \beta)$ and $\Gamma(z + 1) = z\Gamma(z)$.

(a) (1) $\int_0^1 f = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx = \frac{B(\alpha, \beta)}{B(\alpha, \beta)} = 1$ — immediate from the definition of B .

(b) (3) The integrand for $\mathbb{E}[X]$ is the Beta kernel with $\alpha \rightarrow \alpha + 1$:

$$\mathbb{E}[X] = \frac{1}{B(\alpha, \beta)} \int_0^1 x^{\alpha}(1-x)^{\beta-1} dx = \frac{B(\alpha + 1, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha + 1)}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha + \beta + 1)} = \frac{\alpha}{\alpha + \beta}.$$

Marks: 1 for recognising the shifted Beta kernel, 1 for the ratio of B 's, 1 for applying $\Gamma(z + 1) = z\Gamma(z)$ twice to reach $\alpha/(\alpha + \beta)$.

(c) (3) Same move, shifting by two:

$$\mathbb{E}[X^2] = \frac{B(\alpha + 2, \beta)}{B(\alpha, \beta)} = \frac{\Gamma(\alpha + 2)}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha + \beta + 2)} = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)}.$$

Marks: as in (b).

(d) (2)

$$\text{Var}(X) = \frac{\alpha(\alpha + 1)}{(\alpha + \beta)(\alpha + \beta + 1)} - \frac{\alpha^2}{(\alpha + \beta)^2} = \frac{\alpha}{\alpha + \beta} \left[\frac{\alpha + 1}{\alpha + \beta + 1} - \frac{\alpha}{\alpha + \beta} \right] = \boxed{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}}$$

The bracket simplifies because $(\alpha + 1)(\alpha + \beta) - \alpha(\alpha + \beta + 1) = \beta$. Marks: 1 for the subtraction set up, 1 for reaching the single fraction. Award both if they get there by any correct route.

(e) (1) At $\alpha = \beta$ the mean is $\alpha/(2\alpha) = 1/2$. Obvious because $f(x) \propto x^{\alpha-1}(1-x)^{\alpha-1}$ is then symmetric about $x = 1/2$. Marks: 1, requires the symmetry reason, not just the number.

Common errors. Trying to integrate $x^{\alpha}(1-x)^{\beta-1}$ by parts instead of recognising it as another Beta; forgetting the normaliser; algebra slips in (d) — accept any equivalent correct form.

Problem 2 — logit of a uniform (10)

Setup: a completely unknown probability $U \sim \text{Uniform}(0, 1)$ pushed to the log-odds scale, $X = \text{logit}(U)$.

(a) (1) $\sigma(\text{logit}(u)) = \frac{1}{1 + e^{-\log \frac{u}{1-u}}} = \frac{1}{1 + \frac{1-u}{u}} = \frac{u}{u + (1-u)} = u$. Marks: 1.

(b) (3) Since $\log$ is increasing and $u \mapsto u/(1-u)$ is increasing on $(0, 1)$:

$$F_X(x) = \mathbb{P}(\log \frac{U}{1-U} \leq x) = \mathbb{P}(\frac{U}{1-U} \leq e^x) = \mathbb{P}(U \leq \frac{e^x}{1+e^x}) = \mathbb{P}(U \leq \sigma(x)) = \sigma(x),$$

the last step because $\mathbb{P}(U \leq c) = c$ for $c \in (0, 1)$ and $\sigma(x) \in (0, 1)$. *Marks:* 1 for each rearrangement step, 1 for invoking the uniform CDF. Deduct nothing for not remarking on monotonicity, but note it if they invert an inequality carelessly.

(c) (3) $f_X(x) = F'_X(x) = \sigma'(x) = \sigma(x)(1 - \sigma(x))$. Writing $\sigma(x) = 1/(1 + e^{-x})$ and $1 - \sigma(x) = e^{-x}/(1 + e^{-x})$ gives $e^{-x}/(1 + e^{-x})^2$; multiplying numerator and denominator by e^x and using $\cosh x = (e^x + e^{-x})/2$ gives $1/(2(\cosh x + 1))$. It integrates to 1 because $F_X = \sigma$ runs from 0 to 1. *Marks:* 1 for the derivative, 1 for showing the equivalences, 1 for the normalisation argument (the CDF argument is the cleanest; accept direct integration).

(d) (2) $f_X(-x) = \sigma(-x)(1 - \sigma(-x)) = (1 - \sigma(x))\sigma(x) = f_X(x)$, using $\sigma(-x) = 1 - \sigma(x)$. A symmetric density with finite mean has mean at its centre of symmetry, so $\mathbb{E}[X] = 0$. *Marks:* 1 for the symmetry, 1 for the conclusion. They must not just assert mean 0.

(e) (1) The **standard logistic distribution**. It was guaranteed because σ is a CDF, and pushing a $\text{Uniform}(0, 1)$ through the inverse of a CDF produces exactly the law with that CDF — the inverse-transform sampling identity. *Marks:* 1 for the name plus the inverse-transform reason. Name alone: 0.5, round up.

Worth saying in class. $\text{Var}(X) = \pi^2/3 \approx 3.290$; simulation with 4×10^5 draws gave mean $+0.002$ and variance 3.282 . This is the same “sampling is transport” identity the course uses for $Q(U)$ — here the target just happens to be logistic.

Problem 3 — Bernoulli to Binomial, and $\hat{p}$ (10)

Setup: A/B test on a checkout page; n independent visitors, unknown conversion probability p , S conversions.

(a) (1) $\mathbb{P}(x_i = x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$. *Marks:* 1. This single-expression trick is what makes (c) painless.

(b) (3) (i) A specific sequence with k conversions has probability $p^k(1-p)^{n-k}$ by independence — one factor p per conversion and one $(1-p)$ per non-conversion, and the *order* does not change the product. (ii) There are $\binom{n}{k}$ such sequences (choose which of the n visitors converted). (iii) Those sequences are disjoint events whose union is $\{S = k\}$, so their probabilities add. *Marks:* 1 each. Part (iii) is the one usually skipped; insist on it.

(c) (2)

$$L(p) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^S(1-p)^{n-S}, \quad \ell(p) = S \log p + (n-S) \log(1-p).$$

Marks: 1 each.

(d) (3) $\ell'(p) = \frac{S}{p} - \frac{n-S}{1-p} = 0 \Rightarrow S(1-p) = (n-S)p \Rightarrow S = np \Rightarrow \boxed{\hat{p} = S/n}$.

Maximum because $\ell''(p) = -\frac{S}{p^2} - \frac{n-S}{(1-p)^2} < 0$ for all $p \in (0, 1)$ (both terms negative), so ℓ is strictly concave. *Marks:* 1 for the derivative, 1 for solving, 1 for the second-derivative check. The concavity mark is not a formality — do not award it for “it is obviously a max”.

Verified numerically: $n = 200$, $S = 70$; grid search over p maximises at $0.3500 = S/n$.

(e) (1) Nothing changes. The Binomial likelihood is $\binom{n}{S} p^S (1-p)^{n-S}$; the coefficient does not involve p , so it is an additive constant in ℓ and vanishes on differentiation. Same $\hat{p}$. Marks: 1, requires “does not depend on p ”.

Edge cases. If $S = 0$ or $S = n$ the maximiser is at the boundary ($\hat{p} = 0$ or 1) and ℓ' never vanishes in $(0, 1)$. Worth a remark if a student raises it; not required.

Problem 4 — Uniform(a, b) (10)

Setup: second-hand instrument with a lost datasheet; readings uniform on an unknown $[a, b]$.

(a) (3)

$$L(a, b) = \prod_{i=1}^n \frac{1}{b-a} \mathbf{1}\{a \leq x_i \leq b\} = \frac{1}{(b-a)^n} \mathbf{1}\{a \leq x_{(1)}\} \mathbf{1}\{x_{(n)} \leq b\},$$

where $x_{(1)} = \min_i x_i$ and $x_{(n)} = \max_i x_i$. The n indicators collapse into two because *all* observations lie in $[a, b]$ exactly when the smallest and largest do. Marks: 1 for $(b-a)^{-n}$, 1 for including indicators at all, 1 for collapsing them to the min and max. **Omitting the indicator is the central error of this problem** — without it the likelihood appears to increase without bound as $b-a \rightarrow 0$.

(b) (2) Two reasons, either earns the marks if argued: the likelihood is not differentiable in a and b (it jumps to 0 as they cross the data), and where it *is* differentiable the derivative never vanishes — $(b-a)^{-n}$ is strictly decreasing in b and strictly increasing in a , so there is no interior stationary point. The maximum sits on the boundary of the feasible region. Marks: 1 for identifying non-differentiability / boundary maximum, 1 for the “derivative never zero” observation.

(c) (3) On the feasible set $\{a \leq x_{(1)}, b \geq x_{(n)}\}$ the likelihood is $(b-a)^{-n}$, which is maximised by making $b-a$ **as small as possible**. Feasibility forces $b-a \geq x_{(n)} - x_{(1)}$, and that lower bound is attained at $a = x_{(1)}$, $b = x_{(n)}$, which is feasible. Hence $\hat{a} = \min_i x_i$, $\hat{b} = \max_i x_i$. Marks: 1 for “minimise the width”, 1 for the feasibility bound, 1 for checking the bound is attained.

(d) (2) Since $\hat{a} \geq a$ and $\hat{b} \leq b$ always, $\hat{b} - \hat{a} \leq b - a$ with probability 1 — the estimate is **never** too wide and is almost surely too narrow. So it is biased, and biased *downward*. Marks: 1 for “biased”, 1 for the correct direction with the containment reason.

For your own reference (not required of students): for Uniform(a, b), $\mathbb{E}[x_{(1)}] = a + \frac{b-a}{n+1}$ and $\mathbb{E}[x_{(n)}] = b - \frac{b-a}{n+1}$, so $\mathbb{E}[\hat{b} - \hat{a}] = \frac{n-1}{n+1}(b-a)$. Simulated at $a = 2$, $b = 7$, $n = 10$ over 2×10^5 replicates: $\mathbb{E}[\min] = 2.453$ (theory 2.4545), $\mathbb{E}[\max] = 6.547$ (theory 6.5455), width ratio 0.8187 (theory $10/12 = 0.8182$). The unbiased correction multiplies the width by $(n+1)/(n-1)$.

Problem 5 — cross-entropy from maximum likelihood (10)

Setup: postal digit sorter, $K = 10$ classes, logits $z_i = f_\theta(x_i) \in \mathbb{R}^K$, $q_{ik} = e^{z_{ik}} / \sum_m e^{z_{im}}$, labels y_i one-hot.

(a) (2) Positivity: $e^{z_{ik}} > 0$ for every real z_{ik} , and the denominator is a sum of positive terms, so $q_{ik} > 0$. Normalisation:

$$\sum_{k=1}^K q_{ik} = \frac{\sum_{k=1}^K e^{z_{ik}}}{\sum_{m=1}^K e^{z_{im}}} = 1,$$

the numerator and denominator being the same sum. Shift-invariance: replacing z_{ik} by $z_{ik} + c$,

$$\frac{e^{z_{ik}+c}}{\sum_m e^{z_{im}+c}} = \frac{e^c e^{z_{ik}}}{e^c \sum_m e^{z_{im}}} = q_{ik}.$$

So softmax is constant along the all-ones direction: only $K - 1$ of the K logits are genuinely free. The map $\mathbb{R}^K \rightarrow$ (the $(K - 1)$ -simplex) is onto but not injective, which is why implementations may pin $z_K = 0$ without losing anything.

Marks: 1 for positivity **and** normalisation together, 1 for shift-invariance with the $K - 1$ conclusion. Accept “the extra degree of freedom is unidentifiable”.

(b) (2) $y_i \mid x_i \sim \text{Categorical}(q_i)$ — equivalently $\text{Multinomial}(1, q_i)$ — with pmf

$$\mathbb{P}(y_i \mid x_i, \theta) = \prod_{k=1}^K q_{ik}^{y_{ik}}.$$

Because y_i is one-hot, every factor with $y_{ik} = 0$ is $q_{ik}^0 = 1$, so the product collapses to q_{ic_i} , the predicted probability of the true class c_i . This is the exact K -class analogue of $q^y(1 - q)^{1-y}$.

Marks: 1 for naming Categorical/Multinoulli with q_i as parameter, 1 for the single-expression pmf. It must be **conditional on** x_i . Writing q_{ic_i} directly is fine and earns full marks if the one-hot product is also given or clearly implied.

(c) (3)

$$L(\theta) = \prod_{i=1}^n \prod_{k=1}^K q_{ik}^{y_{ik}}, \quad \ell(\theta) = \sum_{i=1}^n \sum_{k=1}^K y_{ik} \log q_{ik}, \quad -\frac{1}{n} \ell(\theta) = \text{CE}(\theta).$$

So **minimising cross-entropy is exactly maximising the likelihood** of the observed labels under the assumed Categorical model; $-1/n$ is a monotone rescaling, so it moves the objective’s value but not its argmax.

The three assumptions:

1. **Form** — each label is Categorical, with its class probabilities given by the model output $\text{softmax}(f_\theta(x_i))$. (Two things bundled: that the label is Categorical at all, and that the softmax link is the right parametrisation.)
2. **Across examples** — the labels are conditionally independent given the inputs, which is what licenses the product over i .
3. **The inputs** — x_i are treated as fixed/given, not modelled. Nothing here assigns a distribution to x , so this is a *conditional* (discriminative) likelihood, not a joint model of (x, y) .

Marks: 1 for the likelihood and log, 1 for landing on CE with the argmax remark, 1 for the assumptions — award the full point only if all three appear. Assumption 3 is the one almost everyone omits and the one worth drawing out in class.

(d) (2) Write the log-softmax in the form that makes the derivative trivial:

$$\log q_k = z_k - \log \sum_m e^{z_m} \implies \frac{\partial \log q_k}{\partial z_j} = \delta_{kj} - \frac{e^{z_j}}{\sum_m e^{z_m}} = \delta_{kj} - q_j.$$

Then

$$\frac{\partial \ell}{\partial z_j} = - \sum_k y_k (\delta_{kj} - q_j) = -y_j + q_j \underbrace{\sum_k y_k}_{=1} = q_j - y_j.$$

The constraint $\sum_k y_k = 1$ is used at exactly one place: pulling q_j out of the sum and collapsing what remains. Without one-hot (or at least sum-to-one) labels the second term would carry a factor $\sum_k y_k \neq 1$.

Marks: 1 for $\partial \log q_k / \partial z_j = \delta_{kj} - q_j$ (however obtained — quotient rule is fine but slower), 1 for the collapse to $q_j - y_j$ **with** the $\sum_k y_k = 1$ step identified. Do not award the second mark if the student writes $q_j - y_j$ without ever using the constraint.

(e) (1) With $K = 2$,

$$q_2 = \frac{e^{z_2}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-(z_2 - z_1)}} = \sigma(z), \quad z := z_2 - z_1,$$

after dividing numerator and denominator by e^{z_2} ; and $q_1 = 1 - \sigma(z)$. Writing $y \in \{0, 1\}$ for the indicator of class 2, the one-hot label is $(1 - y, y)$ and

$$-[(1 - y) \log q_1 + y \log q_2] = -[y \log \sigma(z) + (1 - y) \log(1 - \sigma(z))],$$

which is BCE. Two logits were one too many precisely by the shift-invariance of (a): only the *difference* $z_2 - z_1$ is identifiable.

Marks: 1 for the reduction with the shift-invariance link. Getting $q_2 = \sigma(z_2 - z_1)$ but not connecting it to (a) earns the mark only if the BCE collapse is also shown.

Why this is worth the time. “Predicted minus observed”, now in every coordinate at once, is why softmax classifiers train as easily as they do, and the same structure reappears throughout the course: the two-point posterior $\mathbb{P}(x_0 = +1 \mid x_\sigma) = \sigma(2x/\sigma^2)$ is the $K = 2$ case, and the denoiser’s loss has the same predicted-minus-observed shape. The binary case is derived in the “binary cross-entropy is a likelihood” aside in tutorial 02 — this problem is the general statement that aside is a special case of.

Marking summary

Problem	Topic	Points
1	Beta: mean and variance by integration	10
2	Inverse sigmoid of a uniform $\rightarrow$ logistic	10
3	Bernoulli $\rightarrow$ Binomial; MLE of p	10
4	Uniform(a, b): MLE from the support	10
5	Softmax cross-entropy as maximum likelihood	10
	Total	50

Errors to expect

1. **Omitting the indicator in 4(a)** — the defining error of that problem.
2. Treating 4 as a calculus problem and reporting nonsense stationary points.

3. In 3(b), skipping the disjointness step and just asserting the coefficient.
4. In 5(c), missing that the inputs are unmodelled — the discriminative / generative distinction.
5. In 5(d), producing $q_j - y_j$ without ever invoking $\sum_k y_k = 1$, or grinding through the quotient rule instead of using $\log q_k = z_k - \log \sum_m e^{z_m}$.
6. In 5(a), verifying normalisation but skipping shift-invariance, and so missing that only $K - 1$ logits are free.
7. In 1, attempting integration by parts rather than recognising the Beta kernel.
8. Asserting $\mathbb{E}[X] = 0$ in 2(d) without the symmetry argument.